EKSMA OPTICS

Timing Generator

SY4000

User‘s Manual

2020
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1.1. Warranty Statement

This SY4000 module is protected by one-year warranty covering labor and parts. The warranty enters into validity since the shipment date. Any evidence of improper use or unauthorized attempts at repair leads to warranty cancellation.

1.2. Service Contact Information

In case of service required or any questions on warranty, please notify:

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This document provides SY4000 module specifications and essential features.

The SY4000 can be installed either as optional add-on PCB board, or as a standalone unit. The standalone unit is a SY4000 encased together with a communications module and a power supply.

This document covers interfaces associated with SY4000 module only.

2.1. Short Description

The SY4000 module is a timing generator dedicated to the synchronization of laser components: AOM drivers, Pockels cell drivers, laser diode and flash lamp drivers, detectors, data acquisition systems, etc.

2.1.1. Applications

- Passive or active mode locked, Q-switched lasers, pulsed or QCW
- Data acquisition system triggering
- General purpose pulse generator
- Precise system clock source
- Laser pulse train converting to the clean clock source
- All functions above at once

2.1.2. Highlights

The following are the key features of the SY4000 module in addition to standard pulse generator features:

- Ability to lock to an external clock source, usually photo-diode pulse train. Triggering system is locked to the laser oscillator then, and trigger time is always in phase with the optical pulse.
- Trigger rate control and limiter circuit. It helps to protect laser components from damage due to exceedingly high/low triggering rates.
- Instant switch between two configurations in delay blocks. Burst counter, gate input, frequency divider or software commands may serve as the configuration switching signal sources. Configuration switch is used to control optical pulse pickers (EO or AOM) in a highly flexible manner.
- Low jitter sync pulse output used for high speed acquisition systems like streak camera triggering. Typical jitter is 3...5 ps with respect to the optical pulse.
- Control connector. Software-controllable multiplexer may divert any of the output signals to this connector to sniff what is on other connectors without disturbing them.
- Single ended and differential outputs.
- Clock output: 1:1, 1:2 frequency.
- Up to 4 pulse outputs can be combined to single signals by OR, AND, NOT logical operations.
- DAC output, controlling, e.g. AOM pass though.

2.2. Physical Characteristics

- Board size 100x80x15mm
- Connectors:
    - Front side - BNC
    - Back side - MCX
    - Differential signals - Molex PF-50
- For OEM quantities, boards may be supplied with SMA connectors in front and back.

2.3. Digital Control Interface

The digital control interface is a CAN bus.

Protocol: proprietary Eksma Optics protocol. Protocol description is provided on request.

When SY4000 is paired with a communications module, several different protocols are added, including ASCII text command protocol.

2.4. Specifications

Table 1 Specifications of the SY4000 module

Parameter	***Specification***
Time base	
Internal source	100 MHz 0.2 ppm TCXO
External source, Optical clock	20…100 MHz
Internal rate generator	
Sources	100 MHz clock, Optical clock, SYNC IN, Software command
Rate (T0 period)	50 ns…100 s (0.01 Hz to 20 MHz)
Resolution	10 ns or 1 Optical period or 1 SYNC IN period
RMS jitter	<100 ps
External trigger, SYNC IN input	
System modes	Direct SYNC IN, SYNC IN re-clocked to Optical clock
Rate	DC to 20 MHz
Threshold	1.3 V
Input range	0…5 V
Trigger slope	rising edge
RMS jitter, Direct SYNC IN	<120 ps
RMS jitter, SYNC IN re-clocked	<5 ns
Insertion delay	<80 ns
Delay generators	
Channels, total	8
High res channels	6
High res channels, resolution	25 ps
Low res channels	3
Low res channels, resolution	<10 ns
Delay	0…150 ms
Pulse width	4 ns…150 ms

Parameter	***Specification***
Accuracy, High res channel	2.5 ns + 0.000001 setpoint
Accuracy, Low res channel	≤10 ns + 0.000001 setpoint
Time base, Internal clock	100 MHz, 0.2 PPM TCXO
RMS jitter, channel to channel	<30 ps TTL output, <4 ps PRET output, <20 ps LVDS outputs
Configuration switch	
System modes	Single shot, burst, continuous, frequency divider, GATE IN, inverted GATE IN
Burst counter	1…65535
Frequency divider	1…32767
Outputs, TTL/COMS	
Voltage @50 Ω load	2.5 V or 4.5 V
Voltage @1 MΩ load	5 V or 9 V
Voltage selection 5V/9V	internal jumper
Impedance	50 Ω
Rise time	1.5 ns typ
Output, PRET	
Pulse amplitude @50 Ω load	>1 V
Impedance	50 Ω
Rise time	200 ps typ
Output, differential	
Type	LVPECL
Output, analog	
DAC resolution	12 bit
Max amplitude @50 Ω	1 V
Power	
Voltage	12 V±10%
Power	7 W max

2.5. Input/output Signal Description

2.5.1. Coaxial Connectors, A-side, BNC Type

Table 2 Coaxial connectors, A-side, BNC type

Name	*Description*	*Amplitude*	*Pulse parameters*	*Shape*
SYNC IN	External synchronization input. Trigger or Configuration switch source	LVTTL 0.1 mA pull down to low level	Frequency ≤100 MHz, PW≥10 ns	Positive pulse
PRET OUT	Precise trigger, locked to the optical clock. Delay is adjustable in optical clock period time units. Jitter in respect to the optical pulse is ~4 ps RMS, almost independent on delay.	>1 V on 50 Ω, AC output, capacitor @ output	Rise time ~200 ps, width adjustable >100 ns	Positive pulse
CM TRIG OUT	Common trigger, general purpose output	2.5 V@50 Ω	OUT0 block, programmable pulse parameters	Logic level or pulse with programmable timing
GATE IN	Burst trigger or configuration switch control functions	LVTTL, 0.1 mA pull down to low level	Level or pulse with PW >20 ns	Level or pulse
TRIG1 OUT	General purpose or control sync out	2.5 V@50 om	Select-able multiplexer to OUT0…OUT8 blocks, Optical clock, Optical clock/2	Logic level or pulse with programmable timing

2.5.2. Coaxial Connectors, B-side, MCX Type

Table 3 Coaxial connectors, B-side, MCX type

Name	*Description*	*Amplitude*	*Pulse parameters*	*Shape*
AN OUT	DAC output	0…1 V	N/A	Two programmable analog levels
CM TRIG OUT	Common trigger, general purpose output	2.5 V@50 Ω	OUT0 block	Logic level or pulse with programmable timing
CM TRIG OUT1	Common trigger, general purpose output	2.5 V@50 Ω	OUT0 block	Logic level or pulse with programmable timing
OUT2	General purpose output	2.5 V@50 Ω	OUT2 block, AND+OR+NOT logic with OUT1	Logic level or pulse with programmable timing
OUT3	General purpose output	2.5 V@50 Ω	OUT3 block, AND+OR+NOT logic with OUT2	Logic level or pulse with programmable timing
OUT4	General purpose output	2.5 V@50 Ω	OUT4 block, AND+OR+NOT logic with OUT3	Logic level or pulse with programmable timing

Name	Description	Amplitude	Pulse parameters	Shape
OUT5	General purpose output	2.5 V@50 Ω	OUT5 block, AND+OR+NOT logic with OUT4	Logic level or pulse with programmable timing
CLC IN	External clock input	>100 mV	PW >300ps, input frequency 10..100MHz	Sine, Meander to pulse train from photodiode

Chapter 3 HARDWARE INPUTS & OUTPUTS

3.1. A- & B-side Connectors

Table 4 A-side BNC connectors

Name	Direction	Logic levels	Impedance	Function
GATE IN	Input	LVTTL, tolerates 5V	0.2 mA pull down	Configuration switch and burst control input
SYNC IN	Input	LVTTL, tolerates 5V	0.2 mA pull down	Trigger input
CM TRIG OUT	Output	2.5V @ 50Ω, 4.5V @ 50Ω jumper configurable	50Ω	Master trigger, OUT0 signal
TRIG1 OUT	Output	2.5V @ 50Ω, 4.5V @ 50Ω jumper configurable	50Ω	General trigger, configurable source
PRET OUT	Output	>1.5V @ 50Ω	50Ω	Precision trigger, configurable source

Table 5 B-side MCX connectors

Name	Direction	Logic levels	Impedance	Function
AN OUT	Output	1V @ 50Ω max	50Ω	DAC analog output signal
CM TRIG OUT	Output	2.5V @ 50Ω, 4.5V @ 50Ωjumper configurable	50Ω	Master trigger, OUT0 signal
CM TRIG OUT1	Output	2.5V @ 50Ω, 4.5V @ 50Ωjumper configurable	50Ω	Master trigger, OUT0 signal
OUT2	Output	2.5V @ 50Ω, 4.5V @ 50Ωjumper configurable	50Ω	General purpose trigger, OUT2 signal
OUT3	Output	2.5V @ 50Ω, 4.5V @ 50Ωjumper configurable	50Ω	General purpose trigger, OUT3 signal
OUT4	Output	2.5V @ 50Ω, 4.5V @ 50Ωjumper configurable	50Ω	General purpose trigger, OUT4 signal
OUT5	Output	2.5V @ 50Ω, 4.5V @ 50Ωjumper configurable	50Ω	General purpose trigger, OUT5 signal
CLC IN	Input	0.5V…3.3V pk-pk, sine or pulses	50Ω	Clock input

3.2. Internal Power Connectors

Power 10…24 V.

For pin locations, see Figure 1.

Table 6 Internal power connector pinout

Pin	***Function***	
	Molex MFIT 3.0	***Molex KK 2.5***
1	DC +12…24V	
2	GND	
3	DC +12…24V	
4	GND	

3.3. Internal Signal Connectors

Internal signal connectors are Molex PF-50 type.

Table 7 Internal signal connector pinout

Pin	***Function***			
	X2 CAN+AQC+12V	***X2 CAN***	***X15 DIG OUT1, 12-pin***	***X18 DIG OUT2, 20-pin***
1	GND	GND	GND	GND
2	CANH	CANH	OUT7	OUT1
3	CANL	CANL	OUT7*	OUT1*
4	+12V	+12V	GND	GND
5	GND	GND	OUT8	OUT2
6	ACQ trigger	NC	OUT8*	OUT2*
7	-	-	GND	GND
8	-	-	Reserved for future use	OUT3
9	-	-	Reserved for future use	OUT3*
10	-	-	ACQ_trigger	GND
11	-	-	NC	OUT4
12	-	-	NC	OUT4*
13	-	-	-	GND
14	-	-	-	OUT5
15	-	-	-	OUT5*
16	-	-	-	GND
17	-	-	-	OUT6

Pin	Function			
	X2 CAN+AQC+12V	***X2 CAN***	***X15 DIG OUT1, 12-pin***	***X18 DIG OUT2, 20-pin***
18	-	-	-	OUT6*
19	-	-	-	GND
20	-	-	-	SYNC IN1

3.4. Indicator LEDs

Table 8 Indicator LEDs

Color	***Description***
	DC Power indication
	Is ON when **Command** register = 'ON', 'OFF' in OFF of FAULT modes.
	PLL2 status. Is ON when PLL2 is locked.
	PLL1 status. Is ON when PLL1 is locked. PLL1 is optional. Is ON when 10MHz precision time base module is present. Because of LED intensity is much lower compared to neighbor red LED usually ON state is hardly visible.

3.5. Hardware Jumper Configuration

Hardware jumper locations are marked red in Figure 1.

- J1: CAN terminator
  - CLOSED - CAN inputs are terminated by 120 Ω resistor
  - OPEN - CAN line is not terminated at the board
- J2: A-side output (CM TRIG OUT and TRIG1 OUT) pulse amplitude configuration
  - 1-2 - 2.5 V@50 Ω (5 V@1 MΩ). Recommended
  - 2-3 - repeats B-side output configuration
- J3: B-side output (CM TRIG OUT, CM TRIG OUT1, OUT2, OUT3, OUT4, OUT5) pulse amplitude configuration
  - 1-2 - 4.5V @ 50Ω (9 V@1 MΩ)
  - 2-3 - 2.5V @ 50Ω (5 V@1 MΩ)

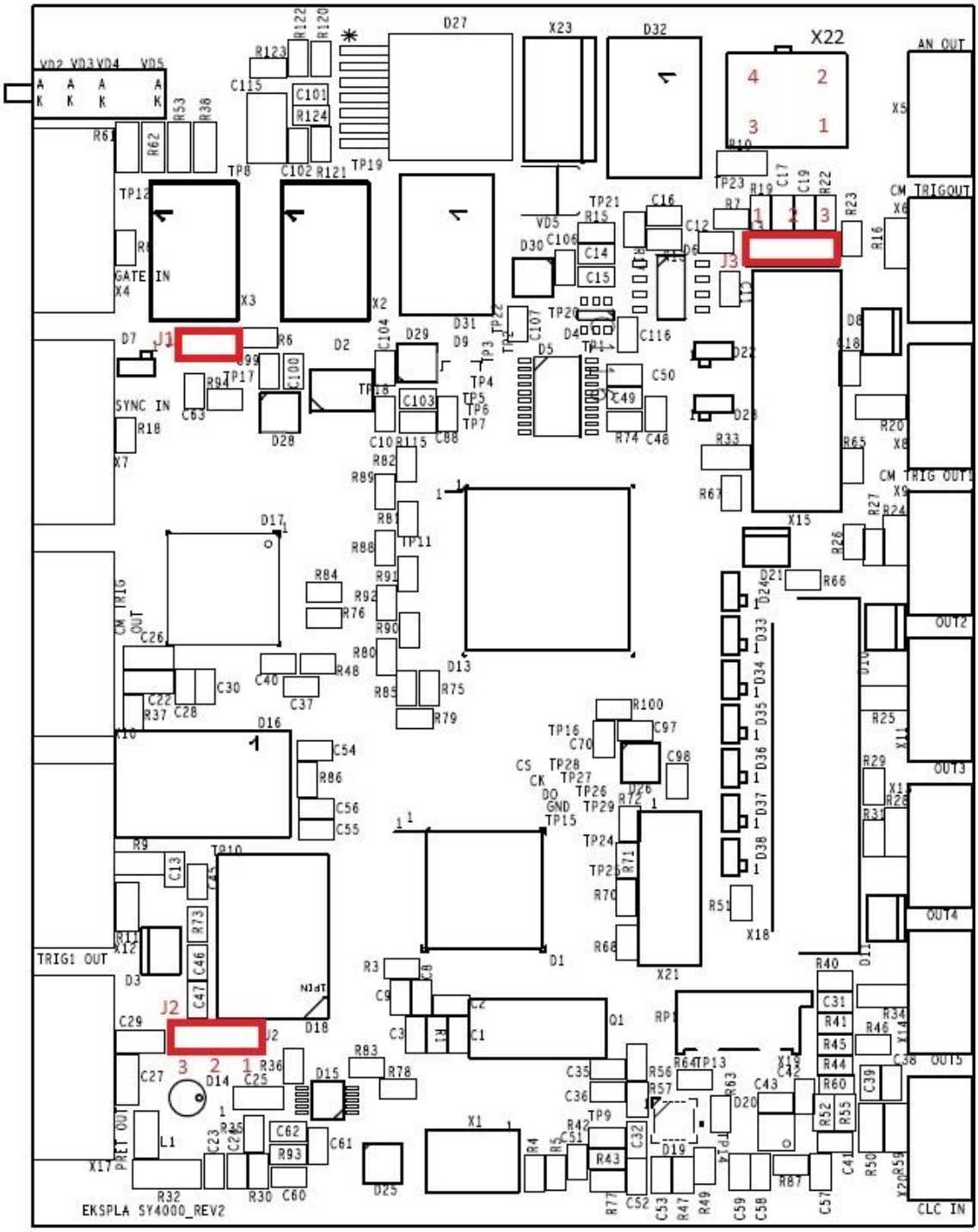


Figure 1 Board drawing

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4.1. Clock System Elements

Clock system diagram is shown in Figure 2.

Table 9 Clock elements

Signal/source name	***Description***	***Frequency***
TCXO	Temperature compensated crystal oscillator, system time base with precision reference option	10MHz, 0.3ppm
VXCO	Voltage controlled crystal oscillator, system time base in standard configuration or VCO in precision reference option	10 x TCXO with precision reference option or 100MHz 20ppm standard
100MHz time base clock	Internal 100MHz clock. In case selected in Optical clock mux serves as time base for System clock of 3GHz. In case external clock selected, 100MHz clocks Measurement counters and Delay counter 4	100MHz, 0.3ppm with precision reference option, 20ppm standard option
OSC OUT	Optical clock mux multiplexer output	100MHz/External source frequency
2.5GHZ, 3GHZ system clock (F)*	F = System clock	OSC OUT * PLL2 N divider , 2920..3080MHz or 2370..2630MHz
F/28	Delay generators time base	System clock divided by 28. ~ 107MHz for rough evaluation
OPT CLOCK	Time base clock in external clock mode, it is cleaned and re-clocked in PLL2 and PLL3 to lower the jitter	CLC IN input frequency

4.2. Clock Control Registers

Table 10 Clock control registers

Type	***Non-volatile***	***Min value***	***Max value***	***Type, dimension, enum values***	***Name***	***Unit or page***	***Value***
u8	NV	0	1	2.5GHz 3GHz	VCO2 frequency	Clock system	From 2.5 GHz to 2370...2630 MHz system clock From 3 GHz to 3000±80 MHz system clock
u8	NV	10	100	%u MHz	Oscillator clock	System clock	Time base frequency rounded to 1MHz
u8	NV	0	1	Internal 100Mhz External	Optical clock mux	System clock	Time base source. Internal 100Mhz - either 100MHz 20ppm source, either 10MHz 0.2ppm x 10 External

Type	*Non-volatile*	*Min value*	*Max value*	*Type, dimension, enum values*	*Name*	*Unit or page*	*Value*
u32	NV	1	18000	%u	PLL2 N divider	System clock	PLL2 N counter. Values allowed: 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 76, 78, 80, 84, 88, 90, 92, 96, 98, 100, 102, 104, 108, 110, 112, 116, 120, 124.

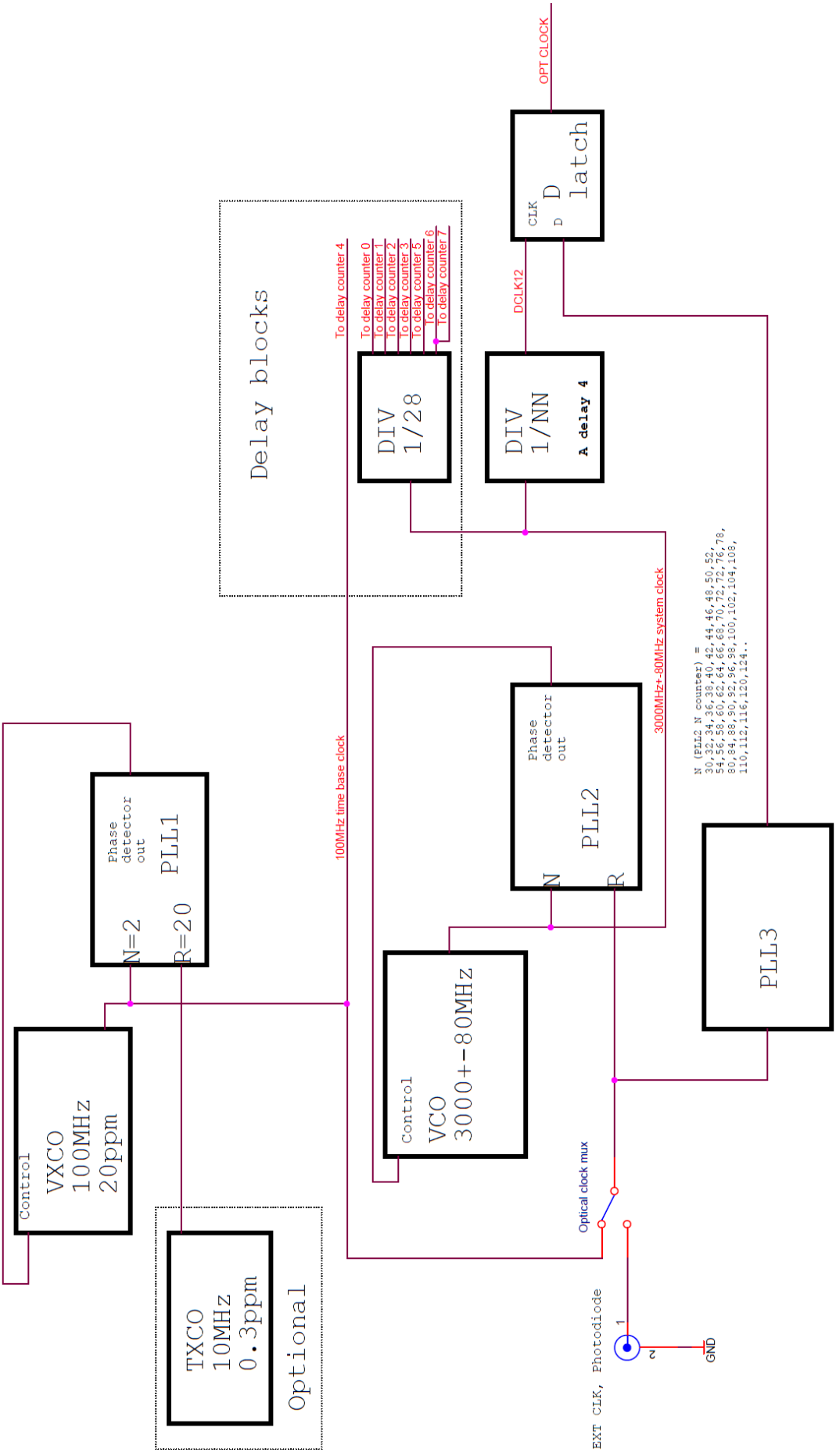


Figure 2 Clock system diagram

The subchapters in this chapter describe the process of SY4000 module configuration step by step.

5.1. Clock System Configuration

5.1.1. Internal 100 MHz Clock

Applications: Q-switched lasers, general triggering system.

Only one possible configuration:

Table 11 Internal 100 MHz clock configuration

Register name	***Value***
Oscillator clock	100 MHz
Optical clock mux	Internal 100 MHz
PLL2 N divider	30

Table 12 Internal 100 MHz clock system characteristics

Signal	***Formula***	***Calculated frequency***	***Calculated period***
System clock (F)		3 GHz	333.333 ps
OPT CLOCK		100 MHz	10 ns
F/28	3 GHz/28	107.1428 MHz	9.333333 ns

5.1.2. External Clock

Applications: mode-locked pico- & femtosecond lasers, multiple synchronization boards locked to a single RF source.

Configuration process:

1. **PLL2 N divider** value is determined. Calculate PLL2 multiplier by dividing 3000 MHz by External clock frequency. Take the closest value from the set of allowed values: 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 76, 78, 80, 84, 88, 90, 92, 96, 98, 100, 102, 104, 108, 110, 112, 116, 120, 124.
2. Check if system frequency is within allowed 2920...3080 MHz range by multiplying External clock frequency by selected multiplier. If frequency is out of range, repeat the step for 2500 MHz system frequency.
3. Set the following registers:
    a. **PLL2 N divider** to value, determined in the first step.
    b. **Oscillator clock** to external clock frequency, rounded to nearest integer value.
    c. VCO2 frequency either to 2.5 GHz or 3 GHz.
    d. **Optical clock mux** to ‘External’.

Table 13 External clock system characteristics

Signal	*Formula*	*Calculated period*
System clock (F)	= **PLL2 N divider** × External clock frequency	1/F
OPT CLOCK	= External clock frequency	1/ (External clock frequency)
F/28	(**PLL2 N divider** × External clock frequency) / 28	= 28 / (**PLL2 N divider** × External clock frequency)

5.2. Triggering System Configuration

The following trigger configurations are common for laser sources in internal trigger mode:

- Internal trigger bound to the internal time base. Mode used for active q-switch lasers or for general pulse generator setup.
- Internal trigger bound to the external clock, used for mode-locked lasers, when trigger with stable phase relations in respect to the optical pulse train is required. Pulse repetition rate is bound to the external frequency, i.e. fixed amount of clock pulses fits in a trigger period.
- Internal trigger bound to the internal 100 MHz clock, but re-clocked to optical clock. Used for mode-locked lasers, when trigger with stable phase relations in respect to the optical pulse train is required together with stable repetition rate. Pulse repetition rate is bound to the internal 100 MHz clock, but trigger keeps stable phase relations in respect to the optical pulse train. This mode features period jitter equal to external clock period (pk-pk), but keeps trigger rate independent on external clock drift.

The following trigger configurations are common for laser sources in external trigger, supplied through 'SYNC IN' pulse, mode:

- External trigger. Straight trigger from SYNC IN pulse rise.
- External trigger re-clocked to optical (external) clock. Trigger event is initiated by SYNC IN pulse that passes D trigger with 'Optical Clock' sent to the clock input. Mode is used when trigger with stable phase relations in respect to the optical pulse train is required.

Trigger rate may be exact SYNC IN frequency or result of dividing by Rate divider.

Triggering system configuration is determined by 3 registers: **Mux Rate divider source**, **Rate divider**, **Mux Trigger**.

Table 14 Common triggering configurations

Triggering mode	*Trigger phase*	*Trigger period*	*Register settings*
Internal	Internal 100 MHz clock	Rate divider × 10ns	**Mux Rate divider source** = 'Internal' **Rate divider** = Required period /10 ns **Mux Trigger** = 'Rate div@OPTCLK'
Internal	External clock	Rate divider × External clock period	**Mux Rate divider source** = 'OPT CLOCK' **Rate divider** = Required period /External clock period **Mux Trigger** = 'Rate div@OPTCLK'
External	External clock	= SYNC IN period	**Mux Trigger** = 'SYNCIN@OPTCLK'

5.3. ON/OFF Control Configuration

ON/OFF control is a feature, specially designed for the laser control. It allows creating two pulse control configurations and alternating them on the fly. Alternating can be done by command of trigger frequency divider (called F DIV), by trigger countdown counter (Burst DIV), and GATE IN input. ON/OFF feature allows

implementing pulse phase or pulse width modulation in addition to plain pulse disabling/enabling. The feature allows burst, gating and pulse repetition control in lasers without, e.g., distorting the thermal lens.

5.3.1. Normal Mode Configuration (Suggestion)

Definitions:

- Configuration 1 - laser is triggered, pump is on, but output pulses are disabled (pulse idling);
- Configuration 2 - laser is firing.

Configuration editing is detailed in 5.4 Pulse Width and Delay Control Configuration.

Manual setting to the idle mode - set register **Mux Conf switch** to '0'. This will set Configuration 1.

Manual stetting to the fire mode - set register **Mux Conf switch** to '1'. This will set Configuration 2.

To allow altering between idle and firing by GATE IN logic input, set register **Mux Conf switch** to 'GATE' or inverted 'nGATE'.

5.3.2. Burst Mode Configuration (Suggestion)

1. Set to burst mode by writing to register **Mux Conf switch** either 'Burst' or 'nBurst'. In 'Burst' active configuration during the burst is '1', in pause between bursts - '0'. 'nBurst' - vice versa.
2. Set the desired number of pulses in the burst by writing 1...65535 to **Burst count** register.
3. Select source of burst trigger by writing to **Mux Burst trigger** register values:
    a. SYNCIN - SYNC IN pulse rise
    b. GATE - GATE IN pulse rise
    c. nGATE - GATE IN pulse fall
    d. One shot burst - software trigger
4. In case 'One shot burst' is selected, write 'START' to **One shot burst** register.

5.3.3. Frequency Divider Configuration (Suggestion)

1. Set to divider mode by writing value 'F Div' to register **Mux Conf switch**.
2. Set desired divider by writing number 1…65535 to register **F div**.

5.4. Pulse Width and Delay Control Configuration

Pulse waveform is generated in **SR blocks**. There are 8 identical 'SR blocks'.

An SR block contains an SR type latch and a chain of multiplexers that allows connecting various signals to S - 'set' and R -'reset' inputs of the latch.

By sending to S and R inputs pulses that are delayed in respect to each other, one may generate a pulse at the block output (nbOUTX).

If the first pulse is sent to S input, a positive logic pulse is obtained, to R - negative.

First multiplexer in the chain is used to switch between configuration 0 and configuration 1, see 5.3 ON/OFF Control Configuration.

The second multiplexer in the chain is used to switch between pulse sources.

To complete a block X configuration, four configurations should be entered:

1. S - (set) for configuration 1, register **Mux set conf 1, block X**.
2. S - (set) for configuration 2, register **Mux set conf 2, block X**.
3. R - (reset) for configuration 1, register **Mux res conf 1, block X**.
4. R - (reset) for configuration 2, register **Mux res conf 2, block X**.

The following signals can be diverted to S and R latch inputs.

> *Note:*
> *The used must make sure to provide S and R pulses that do not overlap in time, because two simultaneously active inputs will send latch to an unpredictable state. Normally this is not hard to do, because all signals, except, 'GATE', 'nGATE' and '1' are of single clock tick width.*

Table 15 S/R latch input signals

Signal	***Description***
0	No state change for SR latch
1	Force set/reset of the latch
Post-count	Channel 0 post counter. See 5.5 Delay Generator Configuration for explanation about post-counters. **Important!** Post-counter of Nth channel is wired to the same Nth block. If a post counter will be used, delay channel and block numbers should match.
Fine delX	Delay generator output, channel X, where X =0...5. Delay channels with 25 ps delay adjustment unit size
Coarse delX	Delay generator output, channel X, where X =6,7 Delay channels with ~9.3 ns unit size and limited capability for smaller units
Trigger	Trigger signal, very first pulse. All delays are calculated in respect to this pulse
Latch SET	Set by software command. Writing of value 'START' to the register **One shot latch set** will set the latch
Latch RES	Reset by software command. Writing of value 'START' to the register **One shot latch reset** will reset the latch
GATE	GATE IN signal
nGATE	Inverted GATE IN signal
Delayed res	Signal taken from R input and delayed for several tens of nanoseconds. It allows generating shortest possible negative logic pulse width at the output
Delayed set	Signal taken from S input and delayed for several tens of nanoseconds. It allows generating shortest possible positive pulse at the output

5.5. Delay Generator Configuration

A possible source of confusion about delay generation in SY4000 is the fact that delays are mostly expressed in counts, not in time units like microseconds or nanoseconds. This is so because the main application is laser products where base clock is not defined in advance and/or may drift in time. In these conditions, the delay must change together with the reference in order to keep a stable phase relationship between triggers and reference.

Delay generator produces pulses with programmable delay. Every delay period starts with the 'Trigger' signal and lasts for a programmable time interval from ~30 ns to ~20 s.

The total delay is a sum of 3 components:

1. **Delay counter**, delaying for fixed amount of F/28 clocks, see Chapter 4 Clock System.
2. **D delay**, adjustable in F clock's.
3. **A delay**, adjustable in 25 ps units.

If an external clock is used, F clock is tied to external reference, therefore expect delay floating together with reference frequency. In this case, delay end phase is fixed in relation with reference clock.

There are 8 delay generators available. Not all generators are identical to each other. Channels 0, 1, 2, 3, 5 are full featured. Channel 4 has a different clock frequency – 100 MHz and lacks the **A delay** and **D Delay** registers. Channels 6 and 7 share the same **A Delay** and **D Delay** register.

General formula for register values calculation is:

*TCount = F/28 * Time*

***Delay Counter** = floor(TCount)*

***D delay** = int((TCount-floor(TCount))*28+4)*

***A delay** = int(((TCount-floor(TCount))*28-floor((TCount-floor(TCount))*28))*13)*

Where:

- Time - delay in seconds
- F/28 - see Chapter 4 Clock System

You may omit **D delay** and **A delay** in case precision greater than ~10 ns is not needed.

Every delay counter has a post-counter connected in cascade. Post counter starts to count after delay counter has reached '0'. This additional delay is used to generate short fixed width pulses. Delay counter (Fine/coarse del X) is connected in this case to 'Set' of SR latch and 'Reset' with Post-counter (Post-countX). Pulse delay is set by pulse generator and pulse width by post-counter. With a single delay generator, it is possible to generate an adjustable width in the range of ~10...2000 ns and delay the pulse. Without post-counter this would require two delay generators.

Channels 1, 2, 3, and 5 have a similar full set of registers. Other channels have a limited set of registers, see Chapter 7 Control Register Summary Table.

Table 16 Delay control registers (channel 0)

Type	***Non-volatile***	***Min value***	***Max value***	***Type***	***Register name***	***Value***
u32	NV	3	16777218	%u	Delay counter 0	Programmable delay in (28/system clock) units: 3…16777218, delay channel 0
u8	NV	4	32	%u	D delay 0	Programmable delay in (1/system clock) units: 4...31, delay channel 0
u8	NV	0	23	%u	A delay 0	Programmable delay in 25 ps units: 0…575 ps, delay channel 0
u16	NV	3	258	%u	Post delay counter 0	Cascade counter, starts when Delay counter finishes count down, channel 0

5.6. Output Configuration

- SR latch block '0' output nbOUT0 is connected directly to CM TRIG OUT connectors: 1pc A-side, 2pcs B-side.
- SR latch block '1' output nbOUT1 is connected to OUT2 MUX multiplexer and differential output OUT1.

- SR latch block '2' through '5' outputs nbOUT2...nbOUT5 are connected to OUT2 MUX...OUT5 MUX multiplexers that allow performing OR, AND, NOT logic operations on nbOUT2...nbOUT5 signals.
- SR latch block '6' and '7' outputs nbOUT6, nbOUT7 are connected to the differential outputs OUT6, OUT7.
- Differential output OUT8 is connected to the Optical clock signal.
- One of multiplexer inputs gets signal from neighboring multiplexer output (exception is OUT2 MUX that receives nbOUT1). One of possible uses of this feature is combining several pulses to the pulse train where delay and with of individual pulses may be adjusted. Up to 5 pulses can be combined in to a train.

A-side connectors TRIG1 OUT and PRET OUT are linked with the outputs from multiplexers MUX TRIG1 OUT and MUX PRET OUT. Multiplexers can divert any nbOUT signal to TRIG1 OUT and PRET OUT connectors. In addition, OPT CLOCK and OPT CLOCK/2 are available at mux inputs. This feature allows watching all outputs without tapping into hardware. Also, TRIG1 and PRET can serve as regular triggering outputs. if single ended outputs are configured for 4.5V @ 50Ω load amplitude, do not select OPT CLOCK as source, because buffer will be overheated and damaged if ran for too long.

PRET OUT connector provides a special type of trigger signal. Before going to the connector, the signal passes D latch, then is clocked by OPT CLOCK signal. Because of the re-clocking, PRET OUT will have a very small jitter (~3ps typical) in respect to external or internal RF source. Combining fast pulse rise time with small jitter and drift, one gets a precision pulse used to trigger high speed measurement devices like streak cameras. Because of re-clocking, fine delay adjustments are lost on PRET OUT - delay and width of pulse can be adjusted in OPT CLOCK units.

5.7. Protection Feature Configuration

5.7.1. Stop on Loss of Lock to Reference Clock Protection

Stop on loss of lock to the reference clock protection is employed automatically when Command = ON is received. Once lock is lost board jumps unconditionally to the FAULT mode. Because all triggering system is tied to the reference clock, output pulses would be unpredictable at loss of lock. Because of this protection, SY4000 cannot be used with an RF source that does not provide a clean clock without amplitude dips and frequency/phase jumps.

5.7.2. Triggering Rate Watchdog

Triggering rate watchdog is intended to protect various components and modules that can be over-stressed, in cases when triggering rate goes over the limits. In laser technology, both types of events (too high and too low triggering rate) may be dangerous, therefore both hazards are treated equally.

Triggering rate may exceed the limits in both triggering modes (internal and external), therefore protection also works in both.

Main hardware part of the triggering rate watchdog is shown in 'Pulse rate limiter' block, see Figure 3

Two actions may be taken when trigger rate is out of limits:

1. Stop triggering and enter the FAULT mode.
2. Forcing or skipping internal trigger pulses in order to keep overall triggering rate within limits.

   Note:
   Trigger rate limiter uses period as a variable. Make sure to covert to period if you are operating in frequency units.

Table 17 Triggering rate watchdog registers

Type	Non-volatile	Min value	Max value	Type, dimension, enum values	Register name	Value
u32	NV	2	2147483647	%u	Trigger Tmin delay	Minimal trigger period in 10 ns units.
u32	NV	2	2147483647	%u	Trigger Tmax delay	Minimal trigger period in 10 ns units.
u8	NV	0	1	OFF ON	Fault on T<Tmin	Hardware action in case of trigger frequency exceeding the limit. ON - fault is tripped OFF - ignored
u8	NV	0	1	OFF ON	Fault on T>Tmax	Hardware action in case of trigger period exceeding the limit. ON - fault is tripped OFF - ignored
u8	NV	0	1	OFF ON	Limit to T<Tmin	Hardware action in case of trigger frequency exceeding the limit. ON - trigger pulse is skipped OFF - ignored
u8	NV	0	1	OFF ON	Limit to T>Tmax	Hardware action in case of trigger period exceeding the limit. ON - trigger pulse is forced OFF - ignored

5.8. Optional Feature Configuration

5.8.1. PHD Divider

PHD divider can be employed when data acquisition, e.g., cannot be triggered at full trigger rate. This block can provide output pulse that is in phase with the trigger, but has a divided repetition rate. PHD divider is controlled by registers.

Table 18 PHD divider registers

Type	Non-volatile	Min value	Max value	Type, enum values	Register name	Value
u8	NV	0	7	nbOUT0 nbOUT1 nbOUT2 nbOUT3 nbOUT4 nbOUT5 nbOUT6 nbOUT7	DAC	Acquisition start trigger source
u16	NV	1	32767	%u	F div Acq	Acquisition trigger divider

5.8.2. DAC

DAC module allows having amplitude modulated pulses at AN OUT connector, B-side of board.

Two amplitude levels are possible to program: one for Configuration 1 and one for Configuration 2.

Amplitude control word is loaded at 'DAC CS' signal rise time.

'DAC CS' pulse must be generated in any of 'SR latch blocks' and has to be diverted by 'MUX DAC CS' multiplexer. DAC is controlled by registers.

Table 19 DAC registers

Type	***Non-volatile***	***Min value***	***Max value***	***Type, enum values***	***Register name***	***Value***
u16	NV	0	4095	%u	DAC1	DAC value in configuration 1
u16	NV	0	4095	%u	DAC2	DAC value in configuration 2
u8	NV	0	7	nbOUT0 nbOUT1 nbOUT2 nbOUT3 nbOUT4 nbOUT5 nbOUT6 nbOUT7	Mux DAC CS	DAC data register load trigger source

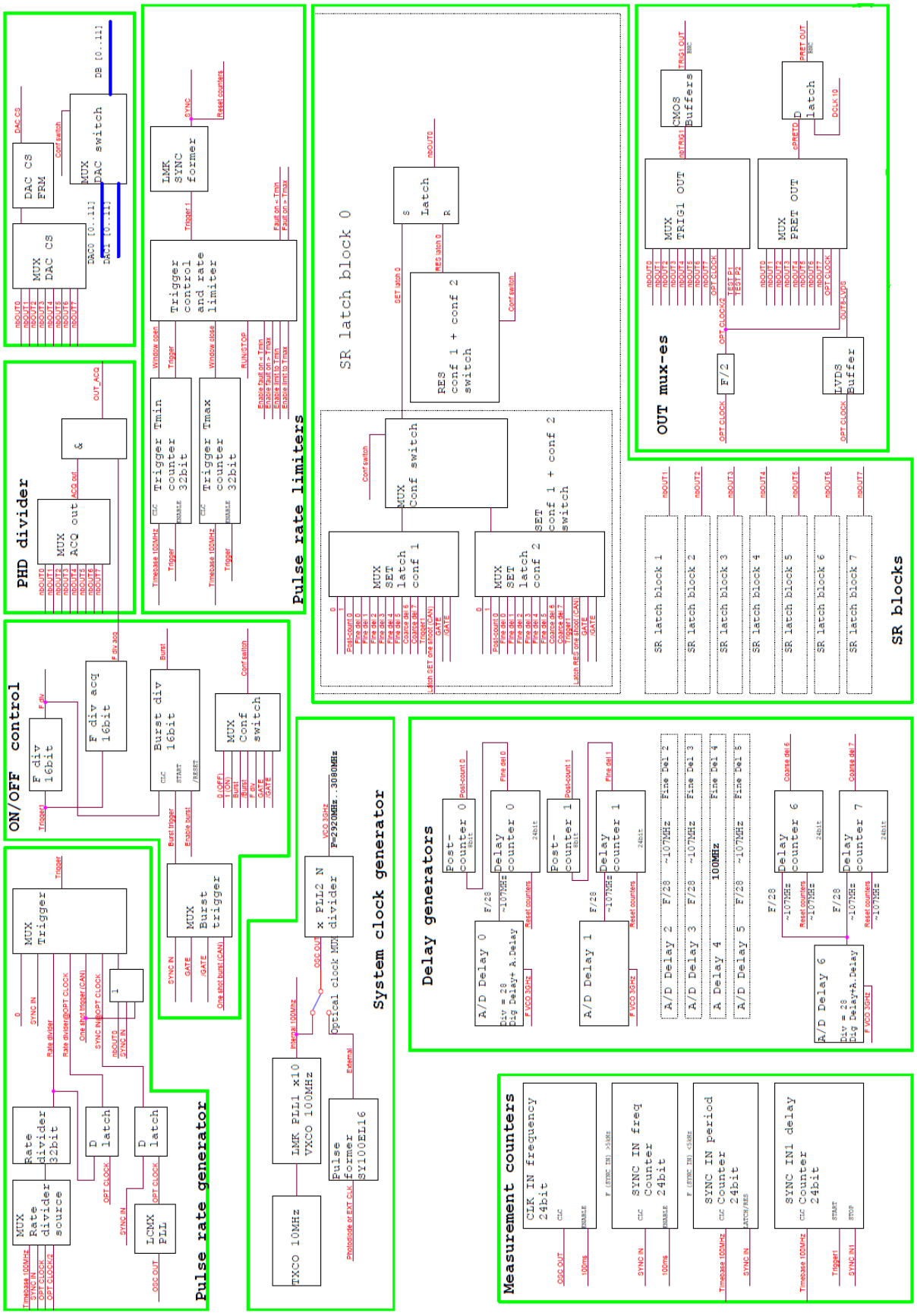


Figure 3 SY4000 functional diagram

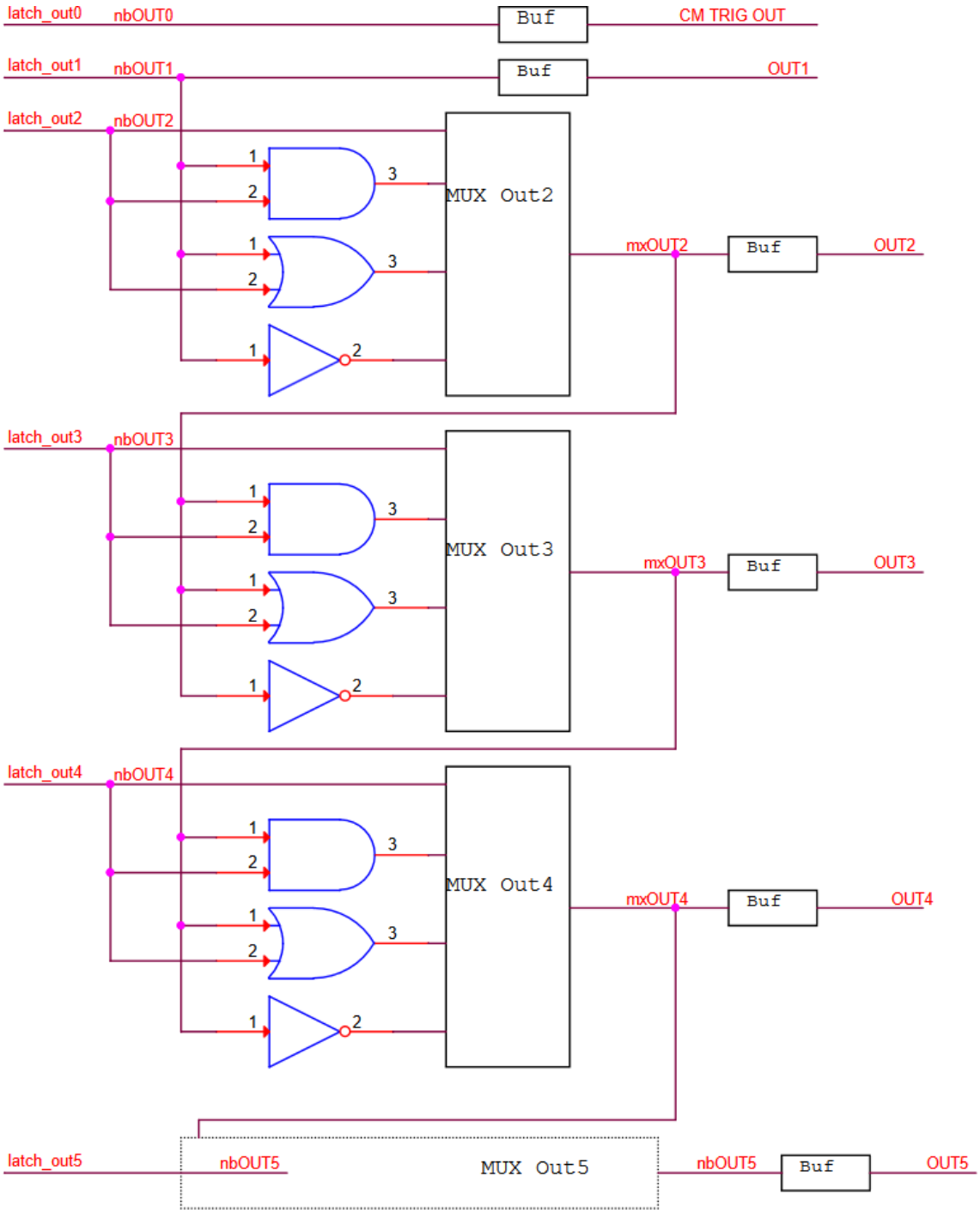


Figure 4 OUT MUX diagram

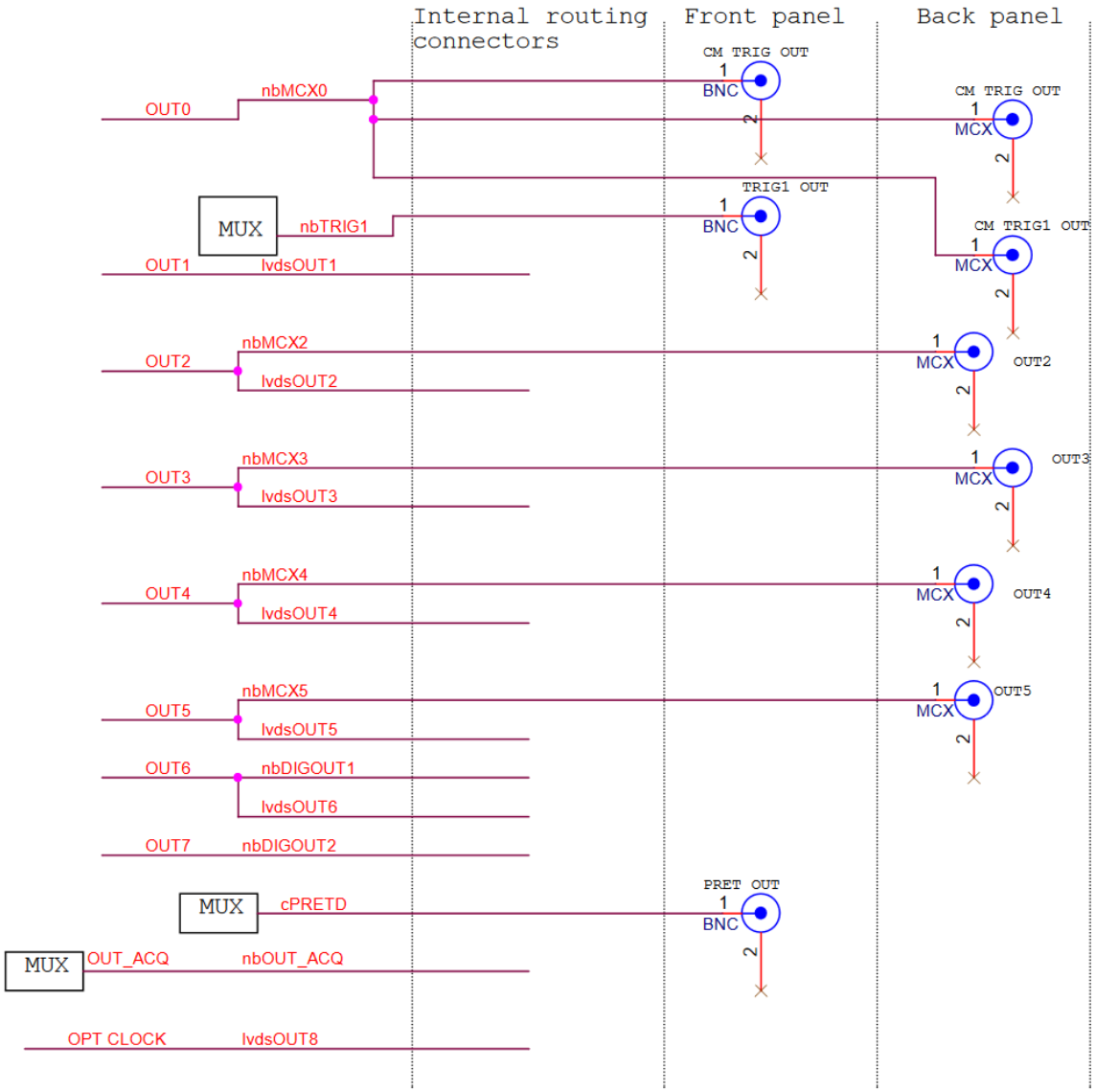


Figure 5 OUTX diagram

Table 20 Control register summary

Type	*Non-volatile*	*Min value*	*Max value*	*Type, dimension, enum values*	*Register name*	*Unit or page*	*Value*
u8		0	1	OFF ON FAULT	Command		**OFF** - No OUT pulses **ON** - Outputs active **FAULT** - FAULT state
u16		0	660	%.2fV	Photodiode bias		
u32	NV	2	2147483647	%u	Trigger Tmin delay	Pulse rate limiter	Minimal trigger period in 10ns units.
u32	NV	2	2147483647	%u	Trigger Tmax delay	Pulse rate limiter	Minimal trigger period in 10ns units.
u16	NV	1	65535	%u	F div	ON/OFF control	Trigger divider
u16	NV	1	32767	%u	F div Acq	ON/OFF control	Acquisition trigger divider
u16	NV	0	4095	%u	DAC1	DAC	DAC value in configuration 1
u16	NV	0	4095	%u	DAC2	DAC	DAC value in configuration 2
u8		0	1	NOLOCK LOCKOK	PLL1 lock	Clock system	10MHz to 100MHz PLL. **NOLOCK** - 10MHz TXCO reference is not present, 100MHz 20ppm source is active **LOCKOK** - 10MHz 0.3ppm reference controls timing system
u8		0	3	NOCLOCK-NOLOCK NOCLOCK-LOCKOK CLOCKOK-NOLOCK CLOCKOK-LOCKOK	PLL2 clock and lock	Clock system	**CLOCK** - wide bandwidth PLL3 status, **LOCK** - narrow bandwidth PLL2 status. **CLOCKOK-LOCKOK** (both loops locked) is required for board to work
u8	NV	0	1	2.5GHz 3GHz	VCO2 frequency	Clock system	2.5GHz - 2370..2630MHz system clock 3GHz - 3000+-80Mhz system clock
s8	NV	-8	8	%d	LagPLL	Clock system	Phase shift between PLL2 and PLL3. Is set during initial test procedure

Type	*Non-volatile*	*Min value*	*Max value*	*Type, dimension, enum values*	*Register name*	*Unit or page*	*Value*
s8		0	18	Logic Low PLL1 DLD PLL2 DLD PLL1 & PLL2 DLD Holdover Status DAC Locked Reserved SPI Readback DAC Rail DAC Low DAC High PLL1_N PLL1_Ndiv2 PLL2_N PLL2_N/2 PLL1_R PLL1_Rdiv2 PLL2_R PLL2_Rdiv2	LD2mux	Hardware	Test pin multiplexer. Used for firmware debugging. Do not change
s8		0	1	- START	One shot trigger	Pulse rate generator block	Software trigger. **Mux trigger** should be set to **One shot trigger**
8		0	1	- START	One shot burst	ON/OFF control block	Software burst trigger. **Mux burst trigger** should be set to **One shot trigger**
s8		0	1	- START	One shot latch set	SR blocks	Software trigger for **set** of **SR latch**. **Mux SET latch** should be set to **Latch SET one shot**
s8		0	1	- START	One shot latch reset	SR blocks	Software trigger for **reset** of **SR latch**. **Mux RES latch** should be set to **Latch RES one shot**
u32	NV	1	65535	%u	Burst count	ON/OFF control	Burst counter initial value
LMK delay block							
u8	NV	0	23	%u	A delay 0	Delay generators	Programmable delay in 25ps units: 0 to 575ps, delay channel 0
u8	NV	0	23	%u	A delay 1	Delay generators	Programmable delay in 25ps units: 0 to 575ps, delay channel 1
u8	NV	0	23	%u	A delay 2	Delay generators	Programmable delay in 25ps units: 0 to 575ps, delay channel 2
u8	NV	0	23	%u	A delay 3	Delay generators	Programmable delay in 25ps units: 0 to 575ps, delay channel 3

Type	Non-volatile	Min value	Max value	Type, dimension, enum values	Register name	Unit or page	Value
u8	NV	0	23	%u	A delay 4	Delay generators	Programmable delay in 25ps units: 0 to 575ps, delay channel 4
u8	NV	0	23	%u	A delay 5	Delay generators	Programmable delay in 25ps units: 0 to 575ps, delay channel 5
u8	NV	0	23	%u	A delay 6	Delay generators	Programmable delay in 25ps units: 0 to 575ps, delay channel 6
u8	NV	4	32	%u	D delay 0	Delay generators	Programmable delay in (1/system clock) units: 4..31, delay channel 0
u8	NV	4	32	%u	D delay 1	Delay generators	Programmable delay in (1/system clock) units: 4..31, delay channel 1
u8	NV	4	32	%u	D delay 2	Delay generators	Programmable delay in (1/system clock) units: 4..31, delay channel 2
u8	NV	4	32	%u	D delay 3	Delay generators	Programmable delay in (1/system clock) units: 4..31, delay channel 3
u8	NV	4	32	%u	D delay 5	Delay generators	Programmable delay in (1/system clock) units: 4..31, delay channel 5
u8	NV	4	32	%u	D delay 6	Delay generators	Programmable delay in (1/system clock) units: 4..31, delay channel 6
Delay counters							
u32	NV	3	16777218	%u	Delay counter 0	Delay generators	Programmable delay in (28/system clock) units: 3..16777218, delay channel 0
u32	NV	3	16777218	%u	Delay counter 1	Delay generators	Programmable delay in (28/system clock) units: 3..16777218, delay channel 1
u32	NV	3	16777218	%u	Delay counter 2	Delay generators	Programmable delay in (28/system clock) units: 3..16777218, delay channel 2
u32	NV	3	16777218	%u	Delay counter 3	Delay generators	Programmable delay in (28/system clock) units: 3..16777218, delay channel 3
u32	NV	3	16777218	%u	Delay counter 4	Delay generators	Programmable delay in 10ns units: 3..16777218, delay channel 4
u32	NV	3	16777218	%u	Delay counter 5	Delay generators	Programmable delay in (28/system clock) units: 3..16777218, delay channel 5
u32	NV	3	16777218	%u	Delay counter 6	Delay generators	Programmable delay in (28/system clock) units: 3..16777218, delay channel 6
u32	NV	3	16777218	%u	Delay counter 7	Delay generators	Programmable delay in (28/system clock) units: 3..16777218, delay channel 7
u16	NV	3	258	%u	Post delay counter 0	Delay generators	Relay race counter, starts when **Delay counter** finishes count down, channel 0
u16	NV	3	258	%u	Post delay counter 1	Delay generators	Relay race counter, starts when **Delay counter** finishes count down, channel 1

Type	Non-volatile	Min value	Max value	Type, dimension, enum values	Register name	Unit or page	Value
u16	NV	3	258	%u	Post delay counter 2	Delay generators	Relay race counter, starts when **Delay counter** finishes count down, channel 2
u16	NV	3	258	%u	Post delay counter 3	Delay generators	Relay race counter, starts when **Delay counter** finishes count down, channel 3
u16	NV	3	258	%u	Post delay counter 4	Delay generators	Relay race counter, starts when **Delay counter** finishes count down, channel 4
u16	NV	3	258	%u	Post delay counter 5	Delay generators	Relay race counter, starts when **Delay counter** finishes count down, channel 5
u16	NV	3	258	%u	Post delay counter 6	Delay generators	Relay race counter, starts when **Delay counter** finishes count down, channel 6
u16	NV	3	258	%u	Post delay counter 7	Delay generators	Relay race counter, starts when **Delay counter** finishes count down, channel 7
Muxes							
u8	NV	0	15	0 1 Post-count Fine del0 Fine del1 Fine del2 Fine del3 Fine del4 Fine del5 Coarse del6 Coarse del7 Trigger Latch SET GATE nGATE Delayed res	Mux set conf, 1 block 0	SR blocks	SR latch is set to '1' state by: **0** - no state change **1** - force to set **Post-count** - channel 0 postcounter **Fine del0** - delay generator, channel 0 **Fine del1** - delay generator, channel 1 **Fine del2** - delay generator, channel 2 **Fine del3** * - delay generator, channel 3 **Fine del4** - delay generator, channel 4 **Fine del5** - delay generator, channel 5 **Coarse del6** - delay generator, channel 6 **Coarse del7** - delay generator, channel 7 **Trigger** - trigger signal **Latch SET** - software command, register 'One shot latch set' **GATE** - GATE IN signal **nGATE** - inverted GATE IN signal **Delayed res** - delayed Reset of the same SR latch, used to generate short pulses

Type	*Non-volatile*	*Min value*	*Max value*	*Type, dimension, enum values*	*Register name*	*Unit or page*	*Value*
u8	NV	0	15	-"-	Mux set conf 2, block 0	SR blocks	SR latch is set to '1' state by: **0** - no state change **1** - force to set **Post-count** - channel 0 postcounter **Fine del0** - delay generator, channel 0 **Fine del1** - delay generator, channel 1 **Fine del2** - delay generator, channel 2 **Fine del3** * - delay generator, channel 3 **Fine del4** - delay generator, channel 4 **Fine del5** - delay generator, channel 5 **Coarse del6** - delay generator, channel 6 **Coarse del7** - delay generator, channel 7 **Trigger** - trigger signal **Latch SET** - software command, register 'One shot latch set' **GATE** - GATE IN signal **nGATE** - inverted GATE IN signal **Delayed res** - delayed Reset of the same SR latch, used to generate short pulses
u8	NV	0	15	-"-	Mux res conf 1, block 0	SR blocks	
u8	NV	0	15	0 1 Post-count Fine del0 Fine del1 Fine del2 Fine del3 Fine del4 Fine del5 Coarse del6 Coarse del7 Trigger Latch RES GATE nGATE Delayed set	Mux res conf 2, block 0	SR blocks	SR latch is reset to '0' state by: **0** - no state change **1** - force to reset **Post-count** - channel 0 postcounter **Fine del0** - delay generator, channel 0 **Fine del1** - delay generator, channel 1 **Fine del2** - delay generator, channel 2 **Fine del3** * - delay generator, channel 3 **Fine del4** - delay generator, channel 4 **Fine del5** - delay generator, channel 5 **Coarse del6** - delay generator, channel 6 **Coarse del7** - delay generator, channel 7 **Trigger** - trigger signal **Latch RES** - software command, register 'One shot latch reset' **GATE** - GATE IN signal **nGATE** - inverted GATE IN signal **Delayed set** - delayed Set of the same SR latch, used to generate short pulses
u8	NV	0	15	-"-	Mux set conf 1, block 1	SR blocks	

Type	Non-volatile	Min value	Max value	Type, dimension, enum values	Register name	Unit or page	Value
u8	NV	0	15	-"-	Mux set conf 2, block 1	SR blocks	
u8	NV	0	15	-"-	Mux res conf 1, block 1	SR blocks	
u8	NV	0	15	-"-	Mux res conf 2, block 1	SR blocks	
u8	NV	0	15	-"-	Mux set conf 1, block 2	SR blocks	
u8	NV	0	15	-"-	Mux set conf 2, block 2	SR blocks	
u8	NV	0	15	-"-	Mux res conf 1, block 2	SR blocks	
u8	NV	0	15	-"-	Mux res conf 2, block 2	SR blocks	
u8	NV	0	15	-"-	Mux set conf 1, block 3	SR blocks	
u8	NV	0	15	-"-	Mux set conf 2, block 3	SR blocks	
u8	NV	0	15	-"-	Mux res conf 1, block 3	SR blocks	
u8	NV	0	15	-"-	Mux res conf 2, block 3	SR blocks	
u8	NV	0	15	-"-	Mux set conf 1, block 4	SR blocks	
u8	NV	0	15	-"-	Mux set conf 2, block 4	SR blocks	
u8	NV	0	15	-"-	Mux res conf 1, block 4	SR blocks	
u8	NV	0	15	-"-	Mux res conf 2, block 4	SR blocks	
u8	NV	0	15	-"-	Mux set conf 1, block 5	SR blocks	
u8	NV	0	15	-"-	Mux set conf 2, block 5	SR blocks	
u8	NV	0	15	-"-	Mux res conf 1, block 5	SR blocks	
u8	NV	0	15	-"-	Mux res conf 2, block 5	SR blocks	
u8	NV	0	15	-"-	Mux set conf 1, block 6	SR blocks	

Type	Non-volatile	Min value	Max value	Type, dimension, enum values	Register name	Unit or page	Value
u8	NV	0	15	-"-	Mux set conf 2, block 6	SR blocks	
u8	NV	0	15	-"-	Mux res conf 1, block 6	SR blocks	
u8	NV	0	15	-"-	Mux res conf 2, block 6	SR blocks	
u8	NV	0	15	-"-	Mux set conf 1, block 7	SR blocks	
u8	NV	0	15	-"-	Mux set conf 2, block 7	SR blocks	
u8	NV	0	15	-"-	Mux res conf 1, block 7	SR blocks	
u8	NV	0	15	-"-	Mux res conf 2, block 7	SR blocks	
u8	NV	0	3	lOUT2 lOUT2 and OUT1 lOUT2 or OUT1 not OUT1	OUT2 mux	OUT multiplexer	See Figure 5 for diagram.
u8	NV	0	3	lOUT3 lOUT3 and OUT2 lOUT3 or OUT2 not OUT2	OUT3 mux	OUT multiplexer	See Figure 5 for diagram.
u8	NV	0	3	lOUT4 lOUT4 and OUT3 lOUT4 or OUT3 not OUT3	OUT4 mux	OUT multiplexer	See Figure 5 for diagram.
u8	NV	0	3	lOUT5 lOUT5 and OUT4 lOUT5 or OUT4 not OUT4	OUT5 mux	OUT multiplexer	See Figure 5 for diagram.

Type	Non-volatile	Min value	Max value	Type, dimension, enum values	Register name	Unit or page	Value
u8	NV	0	15	nbOUT0 nbOUT1 nbOUT2 nbOUT3 nbOUT4 nbOUT5 nbOUT6 nbOUT7 OPTCLK OPCLK div 2 t1 t2 t3 t4 t5 t6	Mux TRIG1 OUT	OUT muxes	TRIG1 OUT output source
u8	NV	0	9	nbOUT0 nbOUT1 nbOUT2 nbOUT3 nbOUT4 nbOUT5 nbOUT6 nbOUT7 OPTCLK OPCLK div 2	Mux PRET OUT	OUT muxes	PRET OUT output source
u8	NV	0	7	nbOUT0 nbOUT1 nbOUT2 nbOUT3 nbOUT4 nbOUT5 nbOUT6 nbOUT7	Mux DAC CS	DAC	DAC data register load trigger source

Type	Non-volatile	Min value	Max value	Type, dimension, enum values	Register name	Unit or page	Value
u8	NV	0	7	nbOUT0 nbOUT1 nbOUT2 nbOUT3 nbOUT4 nbOUT5 nbOUT6 nbOUT7	Mux ACQ out	DAC	Acquisition start trigger source
u8	NV	0	6	0 1 Burst nBurst F div GATE nGATE	Mux Conf switch	ON/OFF control	Configuration switch source. **0** - steady configuration 1 **1** - steady configuration 2 **Burst** - while pause: conf 1, burst count down: conf2 nBurst - inverted *burst* **F div** - while count down: conf 1, overflow - conf 2 **GATE** - GATE IN = '0': Conf 1, GATE IN = '1': Conf2 **nGATE** - inverted GATE
u8	NV	0	3	SYNCIN GATE nGATE One shot burst	Mux Burst trigger	ON/OFF control	Burst trigger source. **SYNCIN** - SYNC IN pulse rise **GATE** - GATE IN pulse rise **nGATE** - GATE IN pulse fall **One shot burst** - software trigger
u8	NV	0	5	Rate divider Rate div@OPTCLK SYNCIN SYNCIN@OPTCLK One shot trig One shot smart	Mux Trigger	Pulse rate generator	Trigger source. **Rate divider** - rate divider **Rate div@OPTCLK** - rate divider reclocked to OPTICAL CLOCK **SYNCIN** - SYNC IN input **SYNCIN@OPTCLK** - SYNC IN input reclocked to OPTICAL CLOCK **One shot trig** - software trigger **One shot smart** - OR function on signals: nbOUT0, SYNC IN and software trigger
u8	NV	0	3	Internal OPTCLK OPTCLKdiv2 SYNCIN	Mux Rate divider source	Pulse rate generator	**Internal** - 100MHz clock **OPTCLK** - OPTCLK internal signal **OPTCLKdiv2** - OPTCLK internal signal divided by 2 **SYNCIN** - SYNC IN input
u32	NV	1	2147483647	%u	Rate divider	Pulse rate generator	Rate divider counter. At =1 rate diver is bypassed
u32	NV	1	18000	%u	PLL2 N divider	System clock	PLL2 N counter. Values allowed: 30,32,34,36,38,40,42,44,46,48,50,52,54,56,58,60,62,64,66,68,70,72,76,78,80,84,88,90,92,96,98,100,102,104,108,110,112,116,120,124.
Setup							
u8	NV	0	1	Internal 100Mhz External	Optical clock mux	System clock	Time base source. **Internal 100Mhz** - either 100MHz 20ppm source, either 10MHz 0.2ppm x 10 **External**

Type	Non-volatile	Min value	Max value	Type, dimension, enum values	Register name	Unit or page	Value
u8	NV	10	100	%uMHz	Oscillator clock	System clock	Time base frequency rounded to 1MHz
u8	NV	0	1	%u	/SYNC Phase	Hardware	
u8	NV	0	1	OFF ON	Fault on T<Tmin	Pulse rate limiter	Hardware action in case of trigger frequency exceeding the limit. **ON** - fault is tripped **OFF** - ignored
u8	NV	0	1	OFF ON	Fault on T>Tmax	Pulse rate limiter	Hardware action in case of trigger period exceeding the limit. **ON** - fault is tripped **OFF** - ignored
u8	NV	0	1	OFF ON	Limit to T<Tmin	Pulse rate limiter	Hardware action in case of trigger frequency exceeding the limit. **ON** - trigger pulse is skipped **OFF** - ignored
u8	NV	0	1	OFF ON	Limit to T>Tmax	Pulse rate limiter	Hardware action in case of trigger period exceeding the limit. **ON** - trigger pulse is forced **OFF** - ignored
u8	NV	0	1	OFF ON	Disable trig-synchronous burst		
u8	NV	0	1	Internal 100Mhz External	Optical clock mux	System clock	Time base source. **Internal 100Mhz** - either 100MHz 20ppm source, either 10MHz 0.2ppm x 10 **External**
Counters							
u32		0	0	%.2fkHz	CLK IN frequency	Measurement counters	Measured **CLC IN** input pulse frequency [kHz]
u32		0	0	%.2fkHz	SYNC IN frequency	Measurement counters	Measured **SYNC IN** input pulse frequency [kHz]
u32		0	0	%u 10ns	SYNC IN period	Measurement counters	Measured **SYNC IN** input pulse frequency [10ns]
u32		0	0	%u 10ns	SYNCIN1 delay	Measurement counters	Measured **SYNC IN1** input pulse rise delay in respect to trigger [10ns]
u32		0	0	%uHz	PLL2 frequency	Measurement counters	
CPLD version							
str8		0	0	%s	CPLD Version string	Hardware	
u32		0	4294967295	%u	CPLD Version number	Hardware	

Type	Non-volatile	Min value	Max value	Type, dimension, enum values	Register name	Unit or page	Value
u32		0	4294967295	%u	CPLD date code	Hardware	
u32		0	0	%0x	LMK id ID_PROD:ID_MASKREV	Hardware	
u8		0	1	%u	Trigger calibrate LMK	Hardware	
u32		0	4294967295	%06u	Firmware SY4000	Hardware	